# Customer Analytics & Churn Prediction

## Project Report

*Big Data Analytics Course*


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# 1. Project Overview

This report presents the design, development, and evaluation of an end-to-end data pipeline for customer analytics and churn prediction. The system integrates large-scale data processing, machine learning, persistent storage, and interactive visualization into a cohesive solution that can support data-driven decision-making in real-world business contexts.

The pipeline is built on three core pillars:

- Apache Spark — for distributed data processing and machine learning at scale
- MongoDB — for flexible, document-oriented data storage
- Streamlit — for building an interactive, web-based analytics dashboard

Together, these technologies form a robust architecture capable of processing over 10,000 customer records efficiently, while providing actionable insights through predictive modeling and visual analytics.

# 2. Project Objectives

The primary objectives of this project are as follows:

- Load and preprocess customer datasets using Apache Spark
- Perform comprehensive data cleaning and feature engineering
- Develop clustering (K-Means) and classification (Logistic Regression) models
- Persist processed data and model outputs in MongoDB
- Deliver an interactive analytics dashboard using Streamlit

# 3. Technologies Used

The project leverages an industry-standard big data stack. The following table summarizes the technologies employed and their respective roles:

| Technology | Role in Project | Version / Library |
|---|---|---|
| Python | Core programming language | 3.x |
| Apache Spark (PySpark) | Distributed data processing & ML | Spark MLlib |
| MongoDB | Document storage & retrieval | Compass + pymongo |
| Pandas & Matplotlib | Data manipulation & charting | Standard libs |
| Streamlit | Interactive web dashboard | Latest stable |
| Scikit-learn / MLlib | Machine learning models | Spark MLlib |

# 4. Dataset Description

The dataset contains comprehensive customer information used to drive analytics and model training. It includes both demographic attributes and behaviorally engineered features that simulate real-world customer engagement patterns.

### 4.1 Core Attributes

- Customer ID — Unique identifier for each customer record
- Name — Full name of the customer
- Country — Geographic location of the customer
- Subscription Date — Date of initial account creation
- Email & Contact Info — Communication details

### 4.2 Behavioral Features (Engineered)

Several features were synthetically generated to simulate customer engagement behavior, enabling the development of meaningful segmentation and predictive models.

## 5. Data Preprocessing

Before modeling, a rigorous preprocessing pipeline was applied to ensure data quality and consistency. The following steps were carried out using Apache Spark:

- Removal of duplicate records to eliminate redundancy
- Imputation and handling of missing values across all feature columns
- Conversion of Subscription Date to a standardized datetime format
- Creation of the tenure_days feature, representing the number of days since subscription

These preprocessing steps ensured that the dataset was clean, consistent, and ready for downstream feature engineering and model development.

## 6. Feature Engineering

Feature engineering was a critical phase in this project, as the raw dataset lacked the behavioral signals necessary for effective churn prediction and customer segmentation. The following features were derived to model customer engagement:

- Activity Score — A composite metric reflecting overall customer engagement
- Monthly Spending — Estimated expenditure per billing cycle
- Login Frequency — Number of platform logins within a defined time window
- Support Tickets — Volume of customer service interactions
- Tenure (Days) — Duration since the customer's initial subscription

These engineered features provided a rich behavioral profile for each customer, significantly improving the discriminative power of the machine learning models.

## 7. Machine Learning Models

### 7.1 K-Means Clustering

K-Means clustering was applied to segment customers into distinct behavioral groups. Segmentation allows the business to tailor strategies for each customer cohort, such as targeted retention campaigns or personalized offers.

- Number of clusters: 3 (Low, Medium, and High engagement segments)
- Features used: Activity Score, Monthly Spending, Login Frequency
- Algorithm: Spark MLlib K-Means with Euclidean distance

### 7.2 Logistic Regression (Churn Prediction)

A binary Logistic Regression classifier was trained to predict customer churn. The model outputs a probability score that is thresholded to produce a binary label.

- Input features: All engineered behavioral features
- Output label: Churn (0 = Retained, 1 = Churned)
- Training framework: Apache Spark MLlib

## 8. Model Evaluation

The Logistic Regression model was evaluated using standard classification metrics to assess its predictive performance on the churn prediction task:

| Metric | Description |
|---|---|
| Accuracy | Proportion of correctly classified instances out of total |
| F1 Score | Harmonic mean of precision and recall; balances false positives and negatives |

Both metrics are well-suited to imbalanced classification tasks such as churn prediction, where the cost of false negatives (missing a churning customer) is typically high.

## 9. MongoDB Integration

Following data processing and model inference, results were persisted in a MongoDB instance to support downstream querying and dashboard consumption. The data flow is as follows:

- Spark DataFrames were converted to Pandas DataFrames for compatibility
- Pandas DataFrames were serialized to Python dictionaries (list of records)
- Records were inserted into the customers collection via the pymongo client
- Database used: myProjectDB | Collection: customers

MongoDB Compass was used throughout development to visually inspect, query, and verify the stored documents, ensuring data integrity at each pipeline stage.

## 10. Streamlit Dashboard

An interactive web-based analytics dashboard was developed using Streamlit, enabling non-technical stakeholders to explore customer data and model outputs without requiring programming knowledge.

### 10.1 Dashboard Features

- Tabular display of the full customer dataset with pagination
- Country-based filtering to drill down into geographic segments
- Churn distribution charts for at-a-glance model output review

- Interactive table view with sortable and searchable columns

The dashboard bridges the gap between raw data pipeline outputs and actionable business insights, making the project's findings accessible to a broad audience.

## 11. End-to-End Project Pipeline

The following table describes each stage of the end-to-end data pipeline, from raw data ingestion through to the interactive analytics dashboard:

| Pipeline Stage | Description |
|---|---|
| **Data Source** | Raw customer CSV dataset ingested into the pipeline |
| **Spark Processing** | Distributed preprocessing and cleaning via PySpark |
| **Feature Engineering** | Behavioral feature derivation for model readiness |
| **ML Models** | K-Means clustering + Logistic Regression churn prediction |
| **MongoDB Storage** | Processed records and model outputs persisted to database |
| **Streamlit Dashboard** | Interactive visualization layer for business users |

## 12. Results & Achievements

The project successfully delivered all planned components. Key outcomes include:

- Successfully processed and analyzed over 10,000 customer records using Spark
- Developed a K-Means clustering model that segmented customers into 3 distinct behavioral cohorts
- Built a Logistic Regression churn prediction model evaluated on Accuracy and F1 Score
- Persisted all processed data and predictions to MongoDB for persistent storage
- Deployed an interactive Streamlit dashboard for data exploration and stakeholder reporting

## 13. Conclusion

This project successfully demonstrates the design and implementation of a complete, production-grade big data analytics pipeline. By integrating Apache Spark for large-scale processing, MongoDB for flexible document storage, and Streamlit for intuitive visualization, the team has built a system that addresses real-world challenges in customer analytics and churn prediction.

The pipeline architecture is modular and extensible, allowing for straightforward upgrades such as real-time data ingestion via Kafka, more sophisticated deep learning models, or deployment as a cloud-native microservices application.

Overall, the project illustrates how modern big data technologies can be combined to deliver meaningful, data-driven insights — a skill set of growing importance in today's data-intensive business landscape.